Haoxin Guo

Urbana, IL | s611guo@gmail.com | linkedin.com/in/haoxin-guo-4a8431351 | github.com/zndwb

Seeking Summer 2027 Internship & Fall 2027 Co-op: Machine Learning Engineer

Education

University of Illinois Urbana-Champaign — Urbana, IL
Master of Computer Science — *Aug 2026 – Expected May 2028*
- Coursework: Advanced Topics in NLP (CS 546), Applied Machine Learning (CS 441), Web Programming (CS 409)

China Agricultural University — Beijing, China
B.E. in Computer Science and Technology; GPA: 3.67/4.00 — *Sep 2022 – Jun 2026*

Internship Experience

Illinois Institute of Technology — Remote
Research Intern (Advisor: Prof. Ren Wang) — *Jul 2025 – Present*
- **MoE unlearning (first author, under review at *IEEE TPAMI*):** Showed that single-expert unlearning in **Mixture-of-Experts** LLMs is superficial (knowledge rebounds after brief fine-tuning) and proposed **PLEU**, a layer-wise expert erasure method. On DeepSeek-V2-Lite and Qwen1.5-MoE, it had **lower knowledge recovery** than single-expert baselines in every setting, preserving MMLU utility with **67–85% less** runtime than mode-connectivity unlearning.
- **Agent unlearning:** Core contributor to a project on unlearning for **self-evolving LLM agents**, removing knowledge stored in agent memory and experience, not only weights; designed the experimental setup and ran experiments.

DP Technology — Beijing, China
Machine Learning Engineer Intern — *Oct 2025 – Jan 2026*
- Pre-trained a **diffusion-based foundation model** from scratch on nanopore (POD5) current signals and re-engineered the PyTorch feature-extraction pipeline, reaching 93% accuracy on core downstream benchmarks.
- Applied **graph representation learning** to RNA–RNA interaction prediction (97% accuracy on in-house data).
- Containerized backend computation with **Docker** on the Bohrium platform and independently shipped a protein normal-mode-analysis web app with automated structure generation and visualization.

Datang Software Technology — Beijing, China
Software Engineer Intern — *Aug 2024 – Nov 2024*
- Developed Django backend services integrating an LLM API (Zhipu GLM) to automate text-generation workflows.
- Built data cleaning and ETL jobs with SQL and Spark, and tuned the platform database for query performance.

Project Experience

University of Illinois Urbana-Champaign — Urbana, IL
CulminationNLI: Event-Completion Reasoning in Multilingual LLMs *(CS 546)* — *Fall 2026 – Present*
- Co-developing an **English–Mandarin NLI benchmark** (∼780 pairs) testing if LLMs tell event occurrence from completion, with labels validated by native speakers.
- Evaluating Qwen, Gemma, and Ministral under zero-shot, definition-aware, few-shot, and CoT prompting, with minimal-pair accuracy and over-/under-inference error analysis; paper planned for submission.

China Agricultural University — Beijing, China
Lightweight Tomato Disease Segmentation *(first author)* — *Jul 2023 – Jan 2025*
- Proposed a **MaxMin-Diffusion attention** mechanism with temporal segmentation, reaching 92% accuracy and 0.91 mIoU (+8 pts over CBAM, +19 over self-attention); deployed on-device on a smartphone.

Deep RL Framework for Automatic Hyperparameter Search *(co-first author)* — *Sep 2023 – May 2025*
- Led design and implementation of **DWIDRL**, framing hyperparameter search as an MDP with progressive search-space attenuation; tuning 5 trajectory-classification models across 4 harvester/tractor scenarios, it beat grid search, Bayesian optimization, PSO, and Q-learning by up to **+21% F1** and **+15% accuracy**.

Publications

- **H. Guo**, Y. Shi, R. Wang. "Beyond Superficial Unlearning: Progressive Layer-wise Expert Erasure in MoE Models." *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)* (IF 20.4), under review.
- **H. Guo**, et al. "Efficient Tomato Disease Detection Using MaxMin-Diffusion Mechanism and Lightweight Techniques." *Plants* (IF 4.7), 14(3):354, 2025. *(First author)*
- W. Zhai*, **H. Guo***, J. Pan, C. Wu. "An Automatic Parameter Search Framework for Agricultural Machinery Trajectory Operation Mode Identification by Deeply Weakly Informed Deep Reinforcement Learning." *Computers and Electronics in Agriculture* (IF 10.3), 253:112172, 2026. *(*Equal contribution)*

Technical Skills

- **Languages:** Python, C/C++, SQL, MATLAB, R, Verilog, Bash
- **ML & LLMs:** PyTorch, Hugging Face, LoRA/PEFT, SFT, RLHF/DPO, MoE, Diffusion Models, Graph Learning, RL
- **Systems & Tools:** Docker, Linux, Git, Django, Spark, LangChain, RAG, LLM Agents

Awards

- Grand Prize, Capital "Challenge Cup"; Grand Prize, "BeiDou Star" National Finals; Honorable Mention, MCM